Session 1. Error-Driven Learning Assignment: Loop Errors

Snippet 1

Issue:

The loop runs infinitely because `i--` decreases `i` instead of increasing it, making it move further away from `10` instead of reaching the condition `i < 10`.

Correction:

```
for (int i = 0; i < 10; i++) {
    System.out.println(i);
}
```

Snippet 2

Issue:

`while (count = 0)` is an assignment, not a condition. This causes a compilation error.

Correction:

```
while (count == 0) {
```

Snippet 3

Issue:

The loop executes once because `num > 0` is false after the first increment (`num = 1`).

Correction:

```
do {
    System.out.println(num);
    num++;
} while (num < 5);
```

Snippet 4

Issue:

The comment suggests the expected output is `1` to `9`, but the loop runs from `1` to `10`.

Correction:

```
for (int i = 1; i < 10; i++) {
    System.out.println(i);
}
```

Snippet 5

Issue:

The loop is supposed to count down but `i++` incorrectly increases `i`, causing an infinite loop.

Correction:

```
for (int i = 10; i >= 0; i--) {
    System.out.println(i);
}
```

Snippet 6

Issue:

Only the first `System.out.println(i);` is inside the loop.
`System.out.println("Done");` is outside.

Correction:

```
for (int i = 0; i < 5; i++) {
    System.out.println(i);
    System.out.println("Done");
}
```

Snippet 7

Issue:

`count` is declared but not initialized, causing a compilation error.

Correction:

```
int count = 0;
while (count < 10) {
    System.out.println(count);
    count++;
}
```

Snippet 8

Issue:

The loop decreases `num`, causing it to exit immediately.

Correction:

```
int num = 5;
do {
    System.out.println(num);
    num--;
} while (num > 0);
```

Snippet 9

Issue:

The loop doesn't run infinitely but prints `0, 2, 4`, which might be unexpected.

Correction:

```
for (int i = 0; i < 5; i++) {
    System.out.println(i);
}
```

Snippet 10

Issue:

`while (num = 10)` is an assignment, not a condition, causing a compilation error.

Correction:

```
while (num == 10) {
```

Snippet 11

Issue:

The loop prints `0, 2, 4` instead of `0, 1, 2, 3, 4` due to `i += 2`.

Correction:

```
while (i < 5) {
    System.out.println(i);
    i++;
}
```

Snippet 12

Issue:

The variable `x` is declared inside the loop, making it inaccessible outside.

Correction:

```
int x = 0;
for (int i = 0; i < 5; i++) {
    x = i * 2;
}
System.out.println(x);
```

SECTION 2: Guess the Output

Snippet 1:

```
public class NestedLoopOutput {

public static void main(String[] args) {

for (int i = 1; i <= 3; i++) {

for (int j = 1; j <= 2; j++) {

System.out.print(i + " " + j + " ");

}

System.out.println();}

}

}
```

Dry Run:

I	J	Output
1	1	11
1	2	12
2	1	21
2	2	22
3	1	31
3	2	32

Expected Output: 1112 (newline) 2122 (newline) 3132

Snippet 2:

```
public class DecrementingLoop {

public static void main(String[] args) {

int total = 0;

for (int i = 5; i > 0; i--) {

total += i;
```

```
if (i == 3) continue;

total -= 1;

}

System.out.println(total);


}
```

Dry Run:

i	total += i	i == 3?	total -= 1	total
5	total = 5	No	total = 4	4
4	total = 8	No	total = 7	7
3	total = 10	Yes (skip total -= 1)	10	
2	total = 12	No	total = 11	11
1	total = 12	No	total = 11	11

Expected Output:

11

Snippet 3:

```
public class WhileLoopBreak {

public static void main(String[] args) {

int count = 0;

while (count < 5) {

System.out.print(count + " ");

count++;

if (count == 3) break;

}

System.out.println(count);


}
```

Dry Run:

`count`	Output	`count++`	`break?`
0	`0`	1	No
1	`1`	2	No
2	`2`	3	Yes (break)

Expected Output:

0 1 2 3

Snippet 4:

```
public class DoWhileLoop {

public static void main(String[] args) {

int i = 1;

do {

System.out.print(i + " ");

i++;

} while (i < 5);

System.out.println(i);

}
```

Dry Run:

`i` before	Output	`i++`	`i < 5?`
1	`1`	2	Yes
2	`2`	3	Yes
3	`3`	4	Yes
4	`4`	5	No (exit loop)

Expected Output:

```
1 2 3 4 5
```

Snippet 5

```
public class ConditionalLoopOutput {
```

```
public static void main(String[] args) {

int num = 1;

for (int i = 1; i <= 4; i++) {

if (i % 2 == 0) {

num += i;

} else {

num -= i;

}

}

System.out.println(num);


}
```

Dry Run:

`i`	Condition (`i % 2 == 0`)	`num` Update	`num` Value
1	No	`num = num - 1`	0
2	Yes	`num = num + 2`	2
3	No	`num = num - 3`	-1
4	Yes	`num = num + 4`	3

Expected Output:

```
3
```

Snippet 6

```
public class IncrementDecrement {

public static void main(String[] args) {

int x = 5;

int y = ++x - x-- + --x + x++;

System.out.println(y);
```

```
}
```

Dry Run:

1. `++x → 6`
2. `x-- → 6 (but x becomes 5)`
3. `--x → 4`
4. `x++ → 4 (but x becomes 5)`

So, `y = 6 - 6 + 4 + 4 = 8`

Expected Output:

```
8
```

Snippet 7

```
public class NestedIncrement {

public static void main(String[] args) {

int a = 10;

int b = 5;

int result = ++a * b-- - --a + b++;

System.out.println(result);


}
```

Dry Run:

1. `++a → 11`
2. `b-- → 5 (but b becomes 4)`
3. `--a → 10`
4. `b++ → 4 (but b becomes 5)`

So, `result = (11 * 5) - 10 + 4 = 55 - 10 + 4 = 49`

Expected Output:

```
49
```

Snippet 8

```
public class LoopIncrement {
```

```
public static void main(String[] args) {

int count = 0;

for (int i = 0; i < 4; i++) {

count += i++ - ++i;

}

System.out.println(count);


}}
```

Dry Run:

`i` before	`i++`	`++i`	`i` after	`count` Update	`count` Value
0	0	2	2	0 - 2	-2
2	2	2	2	-2 - 2	-4

Expected Output:

```
-4
```